

Five Ways to Fool Yourself About Steering a Frozen Model

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Abstract

We measure six mechanisms for changing the interpersonal style of a frozen Qwen3.6-35B-A3B, and score each one twice: on the likelihood of curated reference responses, and on text the model actually generates. The two measurements disagree, and the disagreement is the result. A residual direction trained directly against paired response likelihood drives its training objective from $+0.366$ to -1.188 and decorrelates its axes to a mean cross-cosine of 0.031 , then moves generated text by 5.1% of the explicit-prompt span on warmth and -3.7% on patience, where the sign is reversed. Its signed ordering on held-out prompts is $4/8$ and $2/8$, at or below chance. A static direction and a contextual multi-site variant show smaller versions of the same split, and on one axis the two measurements disagree in sign. The teacher-forced number is the one a steering study most naturally reports, it is measured in the space the writer was optimized in, and on this evidence it certifies nothing. Only the free-generation audit with a wrong-axis control separates a writer that works from one that has learned a continuation discriminator. A seventh mechanism, and the only one in this study whose decision rules were frozen before it ran, distills the zero state of an always-present soft prefix toward the frozen no-prefix model. It nearly buys silence at zero state: median attribution drift falls to 1.1% and 0.4% of the explicit span on warmth and patience, against a $< 10\%$ rule that the previous rung missed at 17.0% and 12.5% . It pays for that silence with the channel it was supposed to preserve, warmth reaching 17.2% of the explicit span against a $\geq 40\%$ rule. Its preregistered verdict is a valid FAIL, and across the three soft-prefix rungs the two quantities trade off monotonically.

1 Introduction

A steering method is usually reported by how far it moves a likelihood. The appeal is obvious: likelihood is cheap, differentiable, and available on held-out data without generating anything. The risk is equally obvious once stated. If the writer was optimized against that likelihood, then reporting it back measures how well the optimizer did its job, not whether the model’s behaviour changed.

This note makes the gap concrete. We build six writers for three signed social style axes, hold the language model frozen throughout, and score every one of them on both quantities: a teacher-forced span over curated positive and negative reference responses, and a generated span recomputed on greedily decoded text under a wrong-axis control. The mechanisms are ordered by how convincing their teacher-forced numbers are. That order does not survive contact with the second measurement.

Status. This is a confirmatory re-run. The ladder’s qualitative outcomes were established in earlier unpublished work, so its doses, layers, and step counts were selected with knowledge of those outcomes. It makes no preregistration claim, and we say so here rather than let the reader assume otherwise. What the run establishes is that the effect reproduces in a clean implementation that shares no code with the original. Stage J (Section 4) is the single exception. It had never been run anywhere, its rules were registered before its first training step, and they did not move afterwards.

2 Setup

The frozen model is Qwen3.6-35B-A3B at revision `995ad96eacd98c81ed38be0c5b274b04031597b0`, loaded in bfloat16 on one NVIDIA RTX PRO 6000. The three axes are warmth versus cold hostility, patience versus

annoyance, and goodwill versus resentment. The corpus is the synthetic 16-case technical-request set with content-matched styled responses, hash 85f6e3599474ad33c37092b583b21ccfc3175fd85b168c65b735c2dd75332346, split six fit, two development, eight held out. The held-out cases include repeated demands, a direct insult, praise, and gratitude.

The two measurements. For a held-out pair, the *teacher-forced span* is the positive-minus-negative reference NLL contrast under a positive intervention minus the same contrast under a negative one. The *generated span* decodes greedily under each sign and recomputes the contrast on the produced text. Both are quoted as a fraction of the span that visible style instructions achieve (Stage A), because a raw likelihood shift has no scale.

Why a wrong-axis control is not optional. The fitted axes are correlated. A large target span means nothing unless it exceeds the shift caused by steering along a different axis, so every generated measurement below reports both.

3 The ladder

Stage A: the denominator. Explicit style instructions give spans of +0.1655 (warmth), +0.1977 (patience), and +0.0737 (goodwill). Goodwill’s upper bound is weak enough that its later fractions should be read with caution; we return to this in the limitations.

Stage B: static directions. One direction per axis, component, and layer, from mean positive-minus-negative activation differences at the prompt frontier over the fit split, at residual and attention outputs of layers 19, 27, and 35.

Stage C: dose and window. A sweep over residual fractions $\{0.005, 0.01, 0.02, 0.04, 0.08\}$ and injection windows of the first token, the first eight, and all response positions, scored teacher-forced. This is where a writer first looks like it works: best spans reach +0.0508 (patience), +0.0465 (goodwill), and +0.0359 (warmth).

Stage D: what the text says. Table 1 recomputes those writers on generated text. Patience and warmth survive in weakened form, at 32.2% and 21.8% of the explicit span with 6/8 signed orderings each. Goodwill does not: 10.9%, 2/8 signed, and specific on a single prompt out of eight.

Stage F: contextual, multi-site. Directions derived from the specific prompt being answered, injected coherently at all six sites under a fixed total norm budget split as $\text{total}/\sqrt{\text{sites}}$. This is a deliberately privileged upper bound rather than a deployable writer. It produces the largest generated spans in the study, and also the study’s cleanest inversion: on goodwill the teacher-forced span is -0.0265 while the generated span is $+0.0594$. The two measurements disagree in sign on the same writer.

Stage G: train the thing directly. One antisymmetric residual direction per axis at layer 35, initialized from Stage B, trained for 72 steps against paired full-response likelihood with a cross-axis cosine penalty. It succeeds at what it was asked to do. The loss falls from $+0.366$ to -1.188 and the mean absolute cross-axis cosine falls to 0.031, so the axes are no longer the single correlated style factor the fitted geometry started from. Then the generated audit returns $+0.0085$ on warmth (5.1%), $+0.0180$ on goodwill (24.5%), and -0.0073 on patience, a reversal. Signed orderings are 4/8, 5/8, and 2/8.

4 Stage J: the rung that was preregistered

Two rungs above Stage G. Everything so far fails, and everything so far was known to fail before this implementation existed. Two mechanisms above Stage G do move generated text, and are reported in the companion soft-prefix study [2] on the same frozen model, the same corpus hash, and the same audit machinery. Stage H trains eight continuous input embeddings per pole and reaches a median generated span of 71.4% of the explicit-instruction span on warmth, with 24/24 signed and 24/24 specific held-out comparisons. Stage I replaces the per-pole banks with one always-present bank, $\text{prefix}(s) = c + \sum_a s_a d_a$, so that a state of zero is a point on

Table 1: Every writer, scored both ways. Teacher-forced spans are measured in the space the writer was optimized in; generated spans are recomputed on greedily decoded text. Percentages are of the Stage A explicit-prompt span for that axis. Signed counts held-out prompts with the intended ordering; specific counts those where the target span also beat the wrong-axis shift.

Stage	Axis	Teacher	Generated	Gen/A	Signed	Specific
D (static)	warmth	+0.0359	+0.0361	21.8%	6/8	5/8
	patience	+0.0508	+0.0636	32.2%	6/8	6/8
	goodwill	+0.0465	+0.0080	10.9%	2/8	1/8
F (contextual)	warmth	+0.0250	+0.1155	69.8%	—	—
	patience	+0.0573	+0.0953	48.2%	—	—
	goodwill	−0.0265	+0.0594	80.7%	—	—
G (trained)	warmth	*	+0.0085	5.1%	4/8	5/8
	patience	*	−0.0073	−3.7%	2/8	2/8
	goodwill	*	+0.0180	24.5%	5/8	5/8

* Stage G optimizes the teacher-forced objective directly; its training loss falls from +0.366 to −1.188, so a held-out teacher-forced span is not an independent measurement of it.

the same tensor path rather than the absence of a prefix. That is the shape any decaying state variable would need, and it costs reach: warmth falls to 42.7%. Worse, the center is not neutral. Trained by cross-entropy against curated neutral responses, it learned some other valid neutral continuation rather than the model’s own. Its median attribution drift is 17.0%, 12.5%, and 33.4% of the explicit span on warmth, patience, and goodwill against a frozen < 10% rule; it never once reproduced the no-prefix answer exactly; and its mean center-to-off word-Jaccard is 0.215, 0.293, and 0.330 across the three seeds.

The mechanism. Stage J keeps Stage I’s architecture unchanged and replaces only the objective that trains the center. Signed events still update the deltas under contrastive paired likelihood. A center event instead samples an on-policy continuation of up to 24 tokens from the center path, then minimizes the mean token-level forward KL to the frozen no-prefix model over the first decoding frontier and every sampled position. The teacher pass carries no prefix and no gradient, and the center’s loss never sees a curated reference response at all. Prefix dropout converts a quarter of the pole events into center events on the same prompt, and eighteen explicit center steps are appended before the schedule is shuffled. The writer this is meant to produce is the base model at zero state and Stage H at the poles.

What was frozen, and when. Stage J was designed in the development tree and never run, which makes it the one rung of this ladder whose rules can honestly precede its data. They were registered before the first Stage J training step was executed anywhere: model and revision, corpus hash, the three seeds, warmth and patience as primary axes with goodwill preregistered as exploratory, the training schedule, the snapshot-selection rule, both decoding protocols, every threshold in Table 2, and the conditions under which the run would be void. The registration states that the runner computes the verdict from those numbers and that whatever it prints is what this paper reports. It printed FAIL with an empty list of invalidating conditions: the three preregistered seeds all completed, on the pinned revision and corpus, with positive explicit spans in every gated mode and axis.

The center went quiet. Gate 1 nearly passes, and this is the part of Stage J that worked. Attribution drift at zero state falls by more than an order of magnitude from Stage I, to 1.1% and 0.4% of the explicit span under greedy decoding and 4.8% and 1.9% under sampling, all four comfortably inside a rule Stage I failed on every axis. The gate is missed on one clause only: median center-to-off word-Jaccard is 0.565 against a ≥ 0.60 rule. The distilled center is behaviorally near-silent and still not textually identical — it writes a different answer that carries almost the same social attribution.

The signed channel went with it. Gate 2 is not close. Under greedy decoding the warmth relative span is 17.2% against $\geq 40\%$ and patience is 17.8% against $\geq 20\%$; under sampling both fall further. Warmth keeps its per-prompt ordering, 21/24 signed and 21/24 beating the wrong-axis shift, so what remains of the channel is real

Table 2: Stage J against its frozen rules. Medians and counts pool three seeds over eight held-out prompts per axis. Drifts and spans are fractions of the explicit-instruction span for that axis and decoding mode. Values that miss their rule are marked †. Where the two primary axes carry different thresholds the rule column reads warmth / patience.

Gate	Quantity	warmth	patience	Rule
1	center drift, greedy	1.1%	0.4%	< 10%
	center drift, sampled	4.8%	1.9%	< 10%
	center-off Jaccard, greedy	0.565†		≥ 0.60
2	relative span, greedy	17.2%†	17.8%†	≥ 40% / ≥ 20%
	signed of 24, greedy	21	18†	≥ 21 / ≥ 21
	beats wrong axis, greedy	21	12†	≥ 21 / ≥ 16
	beats center drift, greedy	19†	19	≥ 21 / ≥ 16
	relative span, sampled	9.3%†	14.5%†	≥ 40% / ≥ 20%
	signed of 24, sampled	18†	15†	≥ 21 / ≥ 21
	beats wrong axis, sampled	15†	12†	≥ 21 / ≥ 16
	beats center drift, sampled	17†	13†	≥ 21 / ≥ 16
3	trajectory signs correct	no†	no†	required
	adjacent inversions of 6	2†	3†	≤ 1

Table 3: The trade-off, measured. Three soft-prefix rungs on the same frozen model, corpus, audit machinery, and three-seed protocol, greedy decoding, all quantities three-seed medians as a fraction of the explicit-instruction span. Stage H has no shared center, so it has no zero state at which to be silent. Stage H and Stage I are from the companion study’s artifact [2].

Stage	Center objective	warmth		patience		Center
		span	drift	span	drift	Jaccard
H	no shared center	71.4%	–	15.6%	–	–
I	cross-entropy to neutral references	42.7%	17.0%	19.5%	12.5%	0.293
J	on-policy KL to the no-prefix model	17.2%	1.1%	17.8%	0.4%	0.565

and simply small. Patience does not, at 18/24 signed against a rule of 21. Gate 3 fails on both axes: the pooled seven-state trajectory has the wrong sign at the moderate negative state on both primary axes and two and three adjacent inversions against a rule of at most one. Goodwill, the preregistered secondary axis, moves the same way and decides nothing: greedy relative span 6.2%, greedy center drift 1.8%.

Reading the curve. Table 3 is the result Stage J was run to get, and it is not the one the design hoped for. On warmth, the axis with enough signal to carry a conclusion, the two quantities move together and monotonically: as the center is pushed from absent to trained-neutral to distilled-silent, drift falls 17.0% → 1.1% and the span it was supposed to leave alone falls 71.4% → 42.7% → 17.2%. Goodwill traces the same shape, 14.7% → 12.6% → 6.2% of span against 33.4% → 1.8% of drift. Patience does not, and we report that plainly: its span reads 15.6%, 19.5%, 17.8% across the three rungs, which is flat inside its own seed spread — Stage H’s three seeds alone range from 1.9% to 16.4%. Patience contributes the drift half of the curve and nothing to the span half, because it was never a strong channel to spend.

What one prefix cannot be. The two properties are competing for the same parameters. Silence at zero state requires the center to sit where the frozen model already is; causal effect at nonzero state requires the deltas to move the model off that point far enough to matter, while sharing a norm budget and a projection with the center that is being pinned down. Stage J spends roughly the whole channel to buy roughly the whole silence, and the two rungs below it sit on the line between. On this evidence an eight-token soft prefix with an always-present shared center cannot be simultaneously silent at zero state and causally effective at nonzero state. It is a bound on a family of writers, not on a hyperparameter: the failure is not that 66 steps were too few, since all three seeds

selected the final snapshot and dev-split frontier KL was still falling at the end, but that the objective which quiets the center is reading the same parameters the poles need.

Consequence for the read/decay/write loop. We designed this rung as the write path for a decaying session state: a memory that fades toward rest needs a rest state that does nothing and intermediate states that do something proportionate. Stage J measures both halves and neither survives. The rest state is now behaviorally quiet, which is the easy half, and the intermediate states are not ordered — the trajectory audit finds the wrong sign at moderate negative states on both primary axes, so the linear tensor path between a signed event and rest is not a monotone dial even where the endpoints separate. The soft prefix is therefore ruled out as the substrate for that loop by measurement rather than by argument, and a write path that edits the residual stream under an explicit state variable has to be evaluated on its own.

5 What this means for reporting steering

The ranking by teacher-forced evidence and the ranking by behaviour are not the same ranking. Stage G is the most thoroughly optimized writer here, has the cleanest axis geometry, and changes generated text least. Stage F has the weakest teacher-forced numbers of the three and the strongest generated ones, and on one axis its two scores point in opposite directions.

The mechanism is not mysterious. A direction trained against the likelihood of a fixed reference continuation can learn to discriminate that continuation from its content-matched opposite without ever making the model’s own decoding trajectory prefer it. Greedy decoding leaves the trained trajectory at the first token and never returns to it. Nothing in the teacher-forced measurement can see that happen, because the measurement never lets the model choose its own tokens.

This is the same shape as a result we reported for sensor conditioning, where a capacity-matched control reproduced a bridge’s entire bits-per-byte gain while task-aligned probes showed decisive input dependence [1]. Both are instances of one rule, which we now state in the general form the rest of this program keeps rediscovering:

A metric computed in the space an intervention was optimized in certifies nothing. Certification requires an out-of-objective behavioral probe.

In the sensor case the probe is a task-aligned ranking; here it is free generation with a wrong-axis control.

Stage J is the mildest instance we have, which is what makes it worth stating separately. Its center objective is a KL divergence to the frozen model’s own next-token distribution, sampled on the center’s own decoding trajectory — about as close to a behavioral loss as a differentiable objective gets — and it converged, with dev-split frontier KL at the selected snapshot falling to 0.054, 0.063, and 0.010 nats across the three seeds. Read alone, those numbers say the center is the base model. The out-of-objective probe says the center reproduces the base model’s greedy answer exactly on six of 24 held-out comparisons in one seed and on none at all in the other two, and shares 0.565 of its content words. The in-objective number was not wrong. It was measuring the distance the optimizer was told to close, and behavioral identity was a different question that only a behavioral probe was going to answer.

6 Limitations

Goodwill’s explicit-prompt span, +0.0737, is less than half of patience’s, and every goodwill percentage in Table 1 is a fraction of that small denominator. An earlier internal run of the same protocol measured a substantially larger goodwill upper bound, so this axis is unstable across runs and its fractions should be treated as exploratory rather than compared against the other two.

The corpus is synthetic, English, and technical. Eight held-out prompts is a small denominator for counts like 6/8, and this run is single-seed: the companion soft-prefix study found seed-dependent behaviour severe enough that one seed refused where two others complied, so single-seed counts here should be read as indicative. Attribution is scored by the same frozen model under style instructions, which is a likelihood contrast rather than a rubric judgment, but it is not an independent judge; human evaluation would test whether the measured directions agree with reader perception. We report one dose per stage after the Stage C sweep and did not extend Stage G’s budget, so its failure is a statement about 72 steps at this dose, not about the mechanism at every budget.

Stage J is better powered than the rungs below it — three seeds and 24 held-out comparisons per axis and decoding mode, rather than one seed and eight — and it is still small, and its counts are still counts out of 24. It tests one prefix length, one KL direction, and one schedule of roughly 66 steps. The trade-off in Table 3 is three points on one model and one corpus, and its two well-behaved axes are warmth and goodwill; patience is flat across all three rungs and supports the drift column only. What Stage J bounds is the family of always-present shared-center soft prefixes trained this way, not soft prefixes in general, and its FAIL is the verdict its own frozen rules produced rather than a judgment that the mechanism is worthless: Gate 1 came within one clause of passing.

7 Reproducibility

The runner is self-contained. It pins the model revision, the corpus hash, the stage configuration, seeds, and decoding, and writes every number in this note into one summary artifact. The reference run used one NVIDIA RTX PRO 6000, peaked at 66.287 GiB reserved, and completed without an out-of-memory event. Free memory gates are parameterized rather than hard preflight aborts, so the suite is portable to a different card; a 2B smoke path exercises the same code for anyone without an 80 GB-class GPU.

```
uv sync
uv run expression-ladder check
uv run expression-ladder reproduce --suite full --device cuda \
  --output-dir results/ladder-35b
uv run python -m expression_ladder.report results/ladder-35b/summary.json
```

Stage J runs standalone, trains its own initialization, needs nothing from an A–G artifact, and defaults to the three registered seeds. Deviating from them is an invalidating condition the runner detects rather than a flag a reader has to notice. The reference block took 6,402 s on the same card and peaked at 65.674 GiB reserved.

```
uv run expression-ladder stage-j --suite full --device cuda \
  --output-dir results/stage-j-35b
```

The frozen rules are PREREGISTRATION-stage-j.md in the reproduction directory and the threshold dictionary STAGE_J_PREREG in stage_j.py, which a test pins to each other. Every Stage J number in this note is in artifacts/stage-j-35b-summary.json, and the run’s hardware, package versions, and repository commit are in artifacts/stage-j-35b-environment.json.

References

- [1] F. Penov. Conditional LoRA Bridges for Modular Sensor Adaptation of Frozen Small Language Models. Preprint, 2026. <https://github.com/kortexa-ai/legolm.paper>.
- [2] F. Penov. Eight Soft Tokens Can Change a Frozen Model’s Tone. Preprint, 2026. <https://github.com/kortexa-ai/legolm.paper>.